



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Falcon 9 launch data was collected using SpaceX's REST API and Wikipedia web scraping. The data was processed using data wrangling, analyzed with visualizations and SQL, and modeled with classification algorithms to predict successful first-stage landings.
- Collect data from public sources
- Feature selection by EDA
- Predictive ML model to choose impactful characteristics

Introduction

- SpaceX charges \$62 million per Falcon 9 launch with a reusable first stage, compared to competitors charging over \$165 million. Reusability depends on a successful landing.
- Predict whether the Falcon 9 first stage will land successfully based on variables such as launch site, payload mass, orbit type, and flight number

Section 1

Methodology

Methodology

Executive Summary

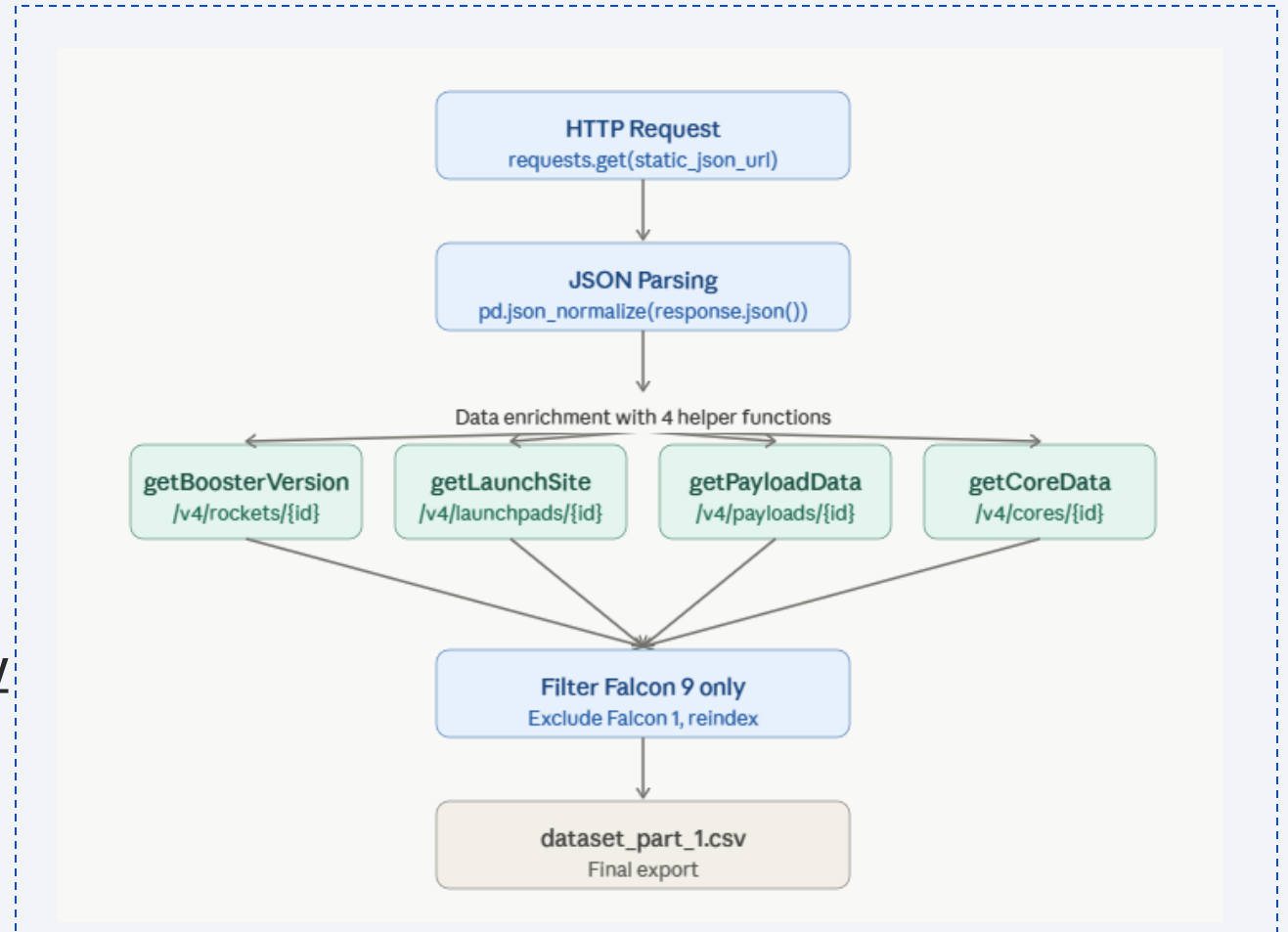
- Data collection methodology:
 - SpaceX REST API
 - Wikipedia web scraping
- Perform data wrangling
 - Cleaning, filtering, and creating the target variable
- Perform exploratory data analysis (EDA):
 - Matplotlib/Seaborn visualizations
 - SQL Queries
- Perform interactive visual analytics:
 - Folium
 - Plotly Dash
- Perform predictive analysis using classification models
 - Construction, tuning, and evaluation

Data Collection

- Falcon 9 launch data was obtained from two sources: SpaceX's public REST API (structured data in JSON) and the Wikipedia page on Falcon 9 launches (via web scraping). Both datasets were combined and cleaned to build the final dataset.
- HTTP requests to the SpaceX REST API → JSON parsing → DataFrame
- Downloading HTML from Wikipedia → BeautifulSoup → table extraction → DataFrame
- Both datasets exported to CSV for the next lab

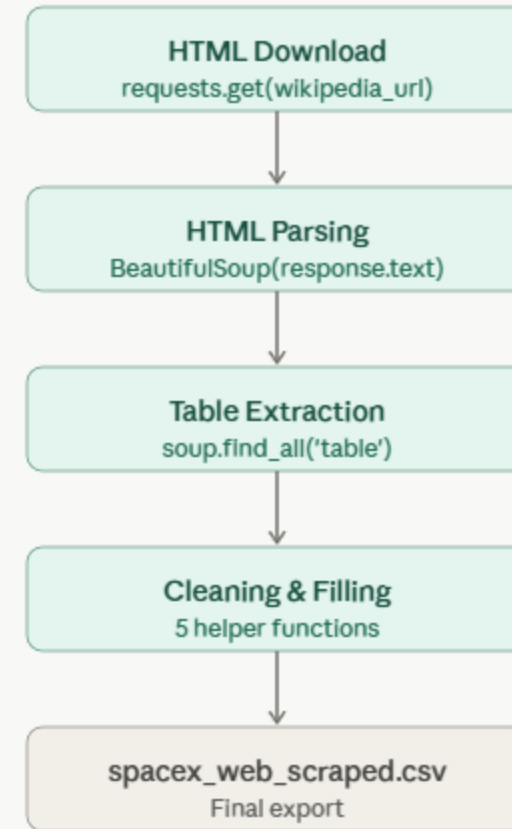
Data Collection – SpaceX API

- HTTP request to the SpaceX REST API → JSON parsing with `pd.json_normalize()` → enrichment with 4 helper functions (booster, site, payload, kernel) → Falcon 9 filtering → export to CSV
- https://github.com/mamemoverflow/testrepo/blob/main/jupyter-labs-spacex-data-collection-api_MC.ipynb



Data Collection - Scraping

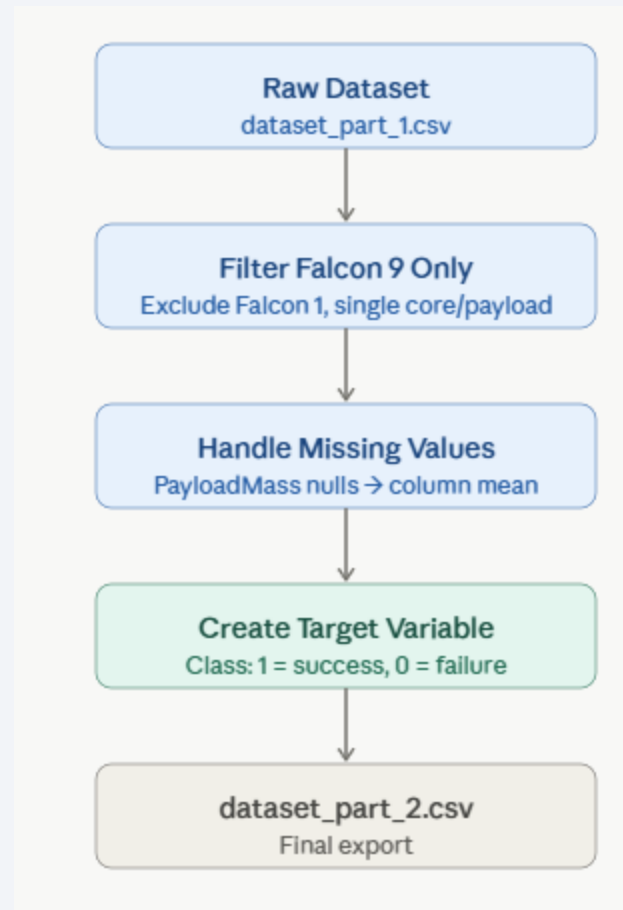
- Download the HTML from Wikipedia using requests, parse it with BeautifulSoup, extract the release table, clean the text with 5 helper functions, and export it to CSV.
- https://github.com/mamemoverflow/testrepo/blob/main/jupyter-labs-webscraping_MC.ipynb



Data Wrangling

- Falcon 1 launches excluded. Rows with multiple cores or payloads removed. Null values in PayloadMass imputed with column mean. Landing outcomes converted to binary target variable Class.
- Filter: keep Falcon 9 only, single core/payload launches
- Target variable Class: 1 = successful landing, 0 = failed
- Imputation: PayloadMass nulls → column mean
- https://github.com/mamemoverflow/testrepo/blob/main/labs-jupyter-spacex-Data%20wrangling_MC.ipynb

Data Wrangling



EDA with Data Visualization

- Flight Number vs Launch Site → identify which sites have more launches and success rate evolution over time
- Payload Mass vs Launch Site → detect payload range per site and its relation to success
- Success rate by Orbit type → identify which orbits have higher landing success
- Flight Number vs Orbit → detect success trends per orbit over time
- Payload Mass vs Orbit → identify payload ranges per orbit and their relation to success
- Annual success rate trend → confirm that success rate improves over the years
- <https://github.com/mamemoverflow/testrepo/blob/main/jupyter-labs-eda-dataviz.ipynb>

EDA with SQL

- Total number of launches per launch site
- Launches with payload mass between 4000 and 6000 kg
- Average payload mass by booster version
- Total successful and failed mission outcomes
- Boosters that carried the maximum payload mass
- Failed drone ship landings in 2015 with booster version and launch site
- Landing outcome counts between 2010-06-04 and 2017-03-20, ranked descending
- https://github.com/mamemoverflow/testrepo/blob/main/jupyter-labs-eda-sql-coursera_sqlite_MC.ipynb

Build an Interactive Map with Folium

- Circle markers for each launch site → visualize geographic location of all 4 sites
- Labels with site names → identify each site directly on the map
- Clustered markers per launch, colored green/red → distinguish successful vs failed landings at a glance
- Distance lines to nearest coastline, city, railway and highway → confirm all sites are close to the coast and transport routes but far from populated areas
- https://github.com/mamemoverflow/testrepo/blob/main/lab_jupyter_launch_site_location_MC.jupyterlite.ipynb

Build a Dashboard with Plotly Dash

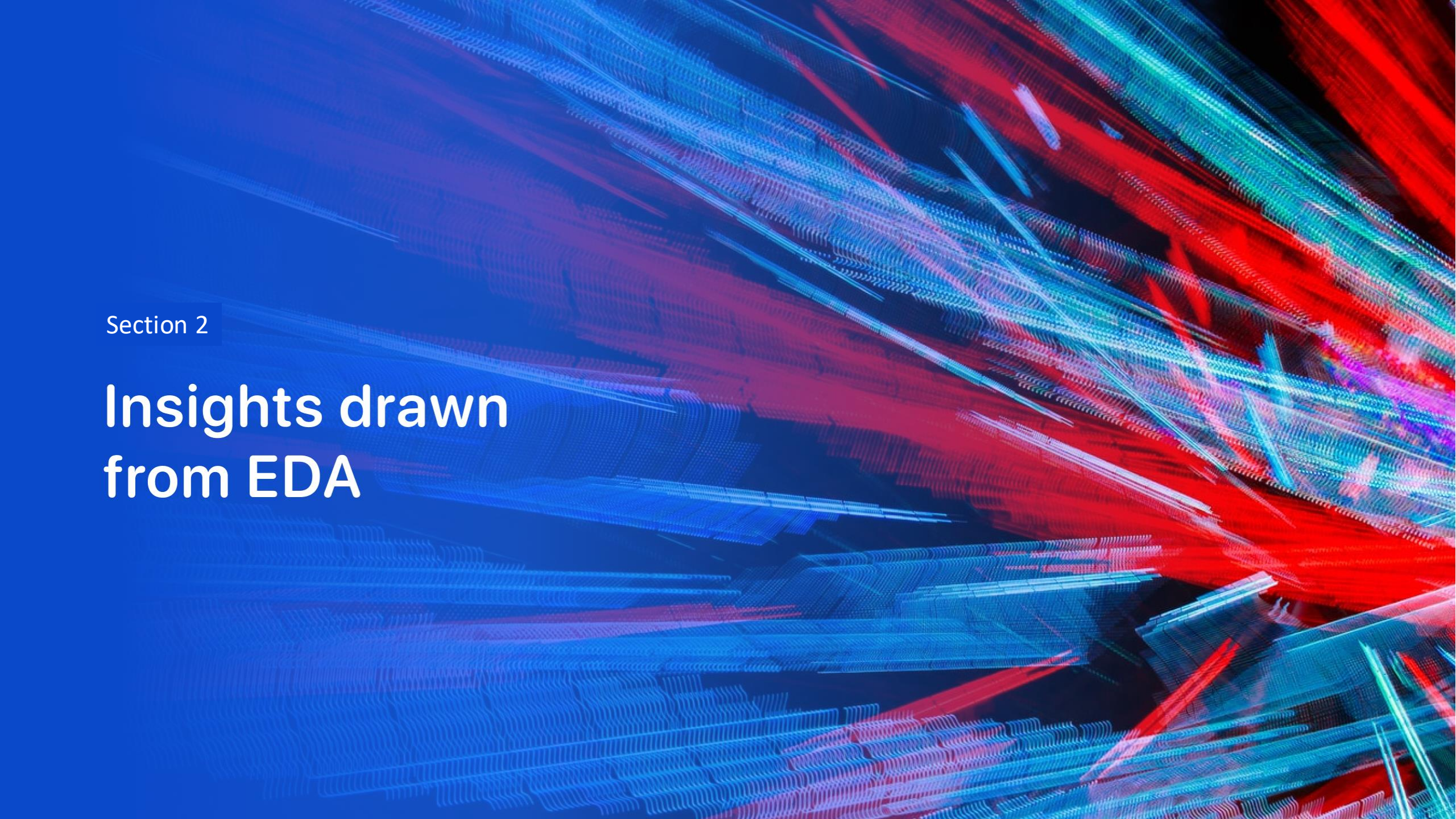
- Interactive dashboard with a dropdown to filter by launch site and a range slider to filter by payload mass. Pie chart shows the proportion of successful launches per site; scatter chart shows the correlation between payload mass and launch success, colored by booster version category.
- Dropdown site-dropdown → filter between all sites or a specific one
- Pie chart → proportion of successful launches per site (or success/failure when a site is selected)
- RangeSlider payload-slider → filter by payload mass range (0-10000 kg)
- Scatter chart → correlation between payload mass and launch success, colored by booster version category
- <https://github.com/mamemoverflow/testrepo/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

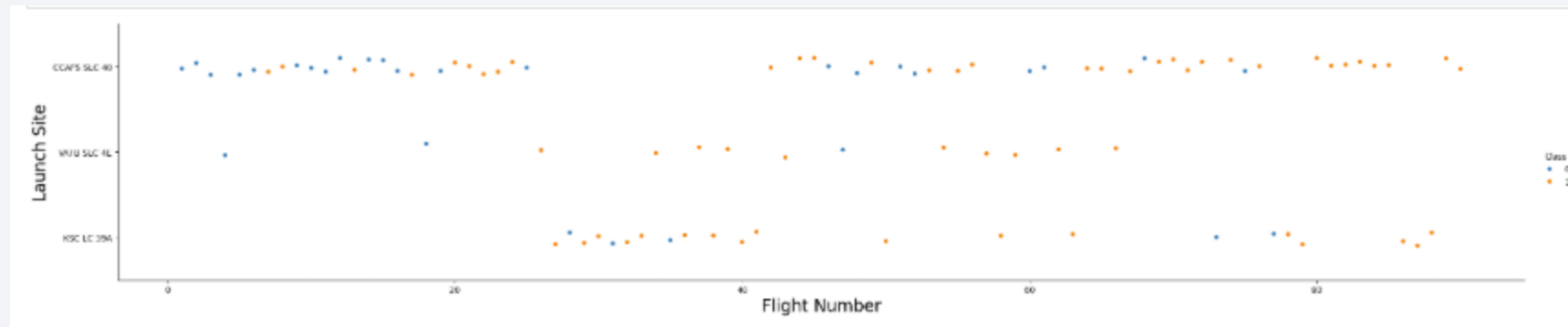
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

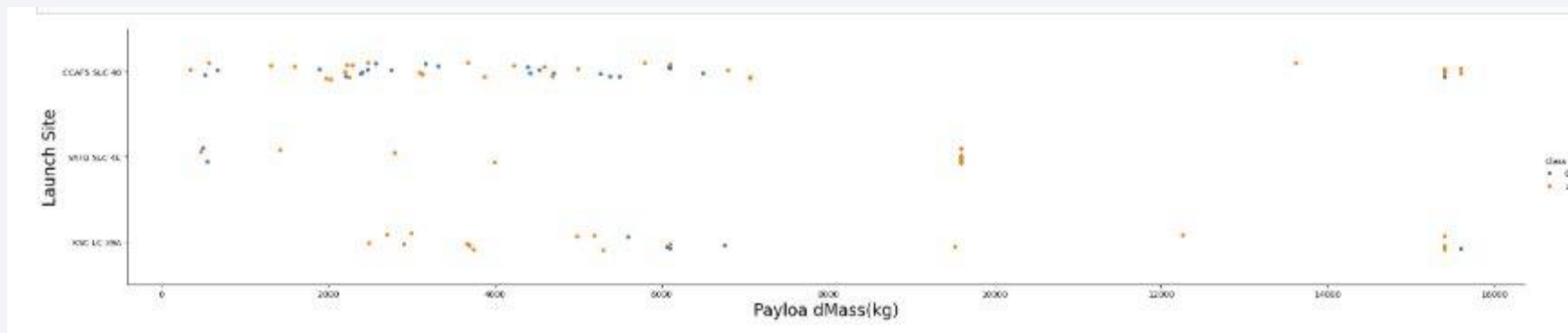
Flight Number vs. Launch Site

- Launch sites CCAFS SLC 40 and KSC LC 39A concentrate the highest number of flights. Success rate (green) increases as flight number grows, suggesting experience improves outcomes.



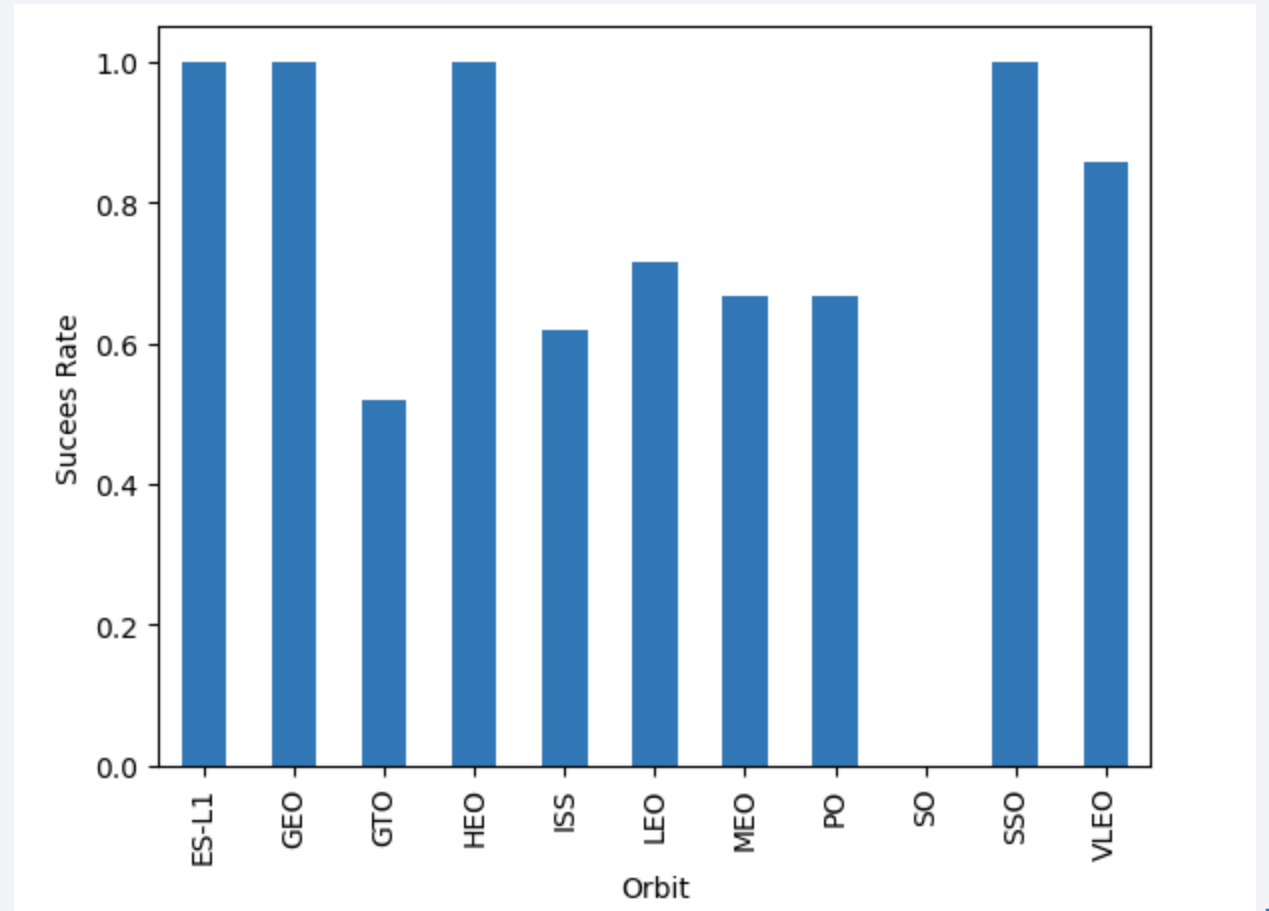
Payload vs. Launch Site

- KSC LC 39A handles the heaviest payloads. CCAFS SLC 40 shows successes across a wide payload range. VAFB SLC 4E is limited to lighter payloads.

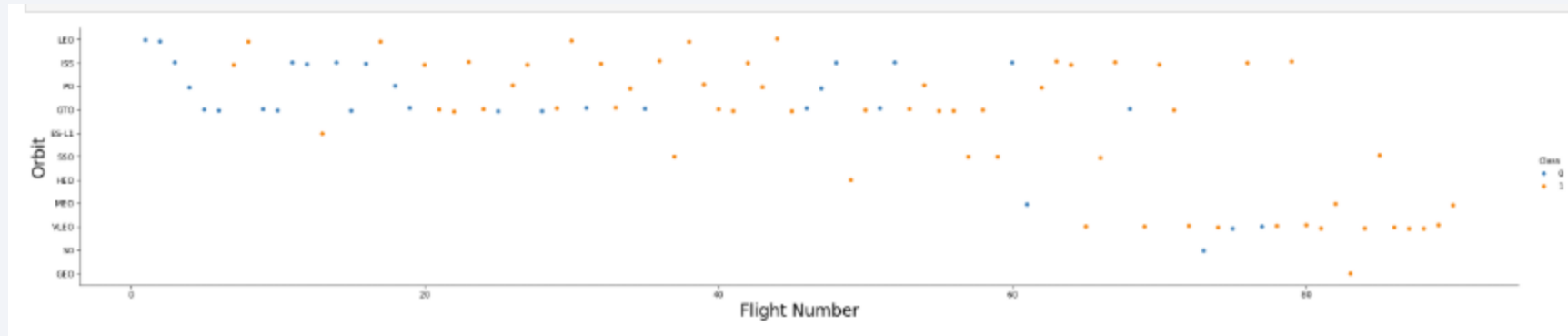


Success Rate vs. Orbit Type

- ES-L1, HEO, SSO and GEO orbits show 100% success rate. LEO and ISS also perform well. GTO shows more mixed results due to higher energy requirements.

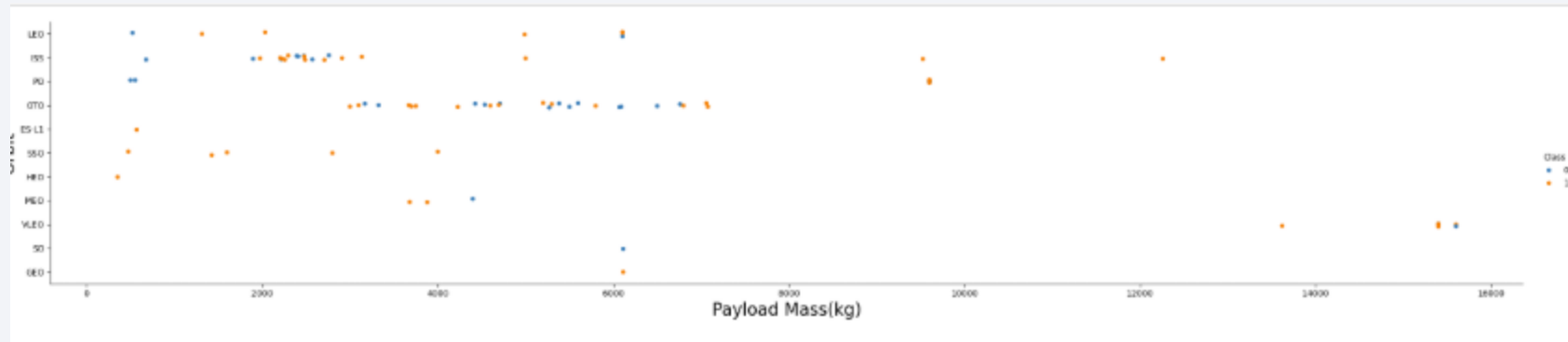


Flight Number vs. Orbit Type



- LEO and ISS missions are distributed across all flight numbers. GTO missions show improving success in later flights. Some orbits like HEO and SSO only appear in recent high-numbered flights.

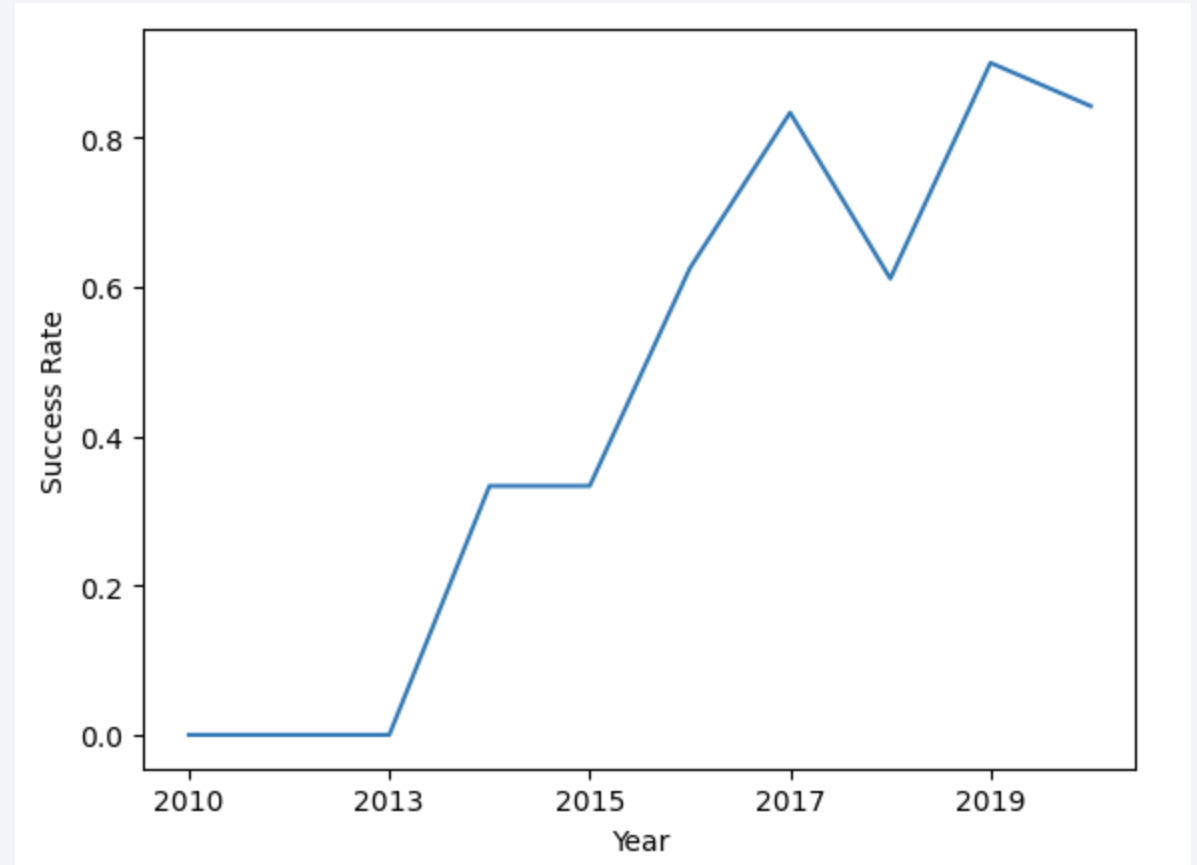
Payload vs. Orbit Type



- GTO missions carry the heaviest payloads. LEO and ISS missions cover a wide payload range. VLEO and SSO are limited to lighter payloads.

Launch Success Yearly Trend

- Success rate was low in 2013-2014, improved significantly from 2015 onwards, and reached near 100% by 2019-2020, reflecting SpaceX's rapid technological progress.



All Launch Site Names

- I used the "DISTINCT" statement to find the names of the unique launch sites
- Query: `%sql select distinct Launch_Site from SPACEXTABLE`
- Query result:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- I used the "WHERE", "LIKE" and "LIMIT 5" statements to find 5 records where launch sites begin with `CCA`
- Query: `%sql select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5`
- Query result:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS __KG__	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- I used the "SUM", "WHERE" and "LIKE" statements to calculate the total payload carried by boosters from NASA.
- Query: `%sql select sum(PAYLOAD_MASS__KG_) from SPACEXTABLE where customer like '%NASA%'`
- Query result:

```
sum(PAYLOAD_MASS__KG_)
```

```
107010
```

Average Payload Mass by F9 v1.1

- I use "AVG" and "WHERE" statements to calculate the average payload mass carried by booster version F9 v1.1
- Query: `%sql select AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE where booster_Version = 'F9 v1.1'`
- Query result:

AVG(PAYLOAD_MASS__KG_)

2928.4

First Successful Ground Landing Date

- I used "MIN", "WHERE" and "NOT IN" to find the dates of the first successful landing outcome on ground pad
- Query: `%sql select min(Date) from SPACEXTABLE where landing_outcome not in ('%failure%', 'no attempts')`
- Query result:

min(Date)
2010-06-04

Successful Drone Ship Landing with Payload between 4000 and 6000

- I used "SELECT Booster_Version", "WHERE" and "AND" statements to list the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Query: `%sql select Booster_Version from SPACEXTABLE where Landing_Outcome= 'Success (drone ship)' AND (PAYLOAD_MASS__KG_ > 4000 and PAYLOAD_MASS__KG_ < 6000)`
- Query result:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- I used "COUNT" and "GROUP BY" statements to calculate the total number of successful and failure mission outcomes
- Query: `%sql select COUNT(*), Mission_Outcome from SPACEXTABLE GROUP BY Mission_Outcome`
- Query result:

COUNT(*)	Mission_Outcome
1	Failure (in flight)
98	Success
1	Success
1	Success (payload status unclear)

Boosters Carried Maximum Payload

- I used "SELECT Booster_Version", "WHERE" and "IN" statements to list the names of the booster which have carried the maximum payload mass
- Query: `%sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS__KG_ in (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE)`
- Query result:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- I used "SUBSTR()", "WHERE" and "AND" statements to list the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Query: `%sql select substr(Date, 6,2) as month, booster_version, launch_site, landing_outcome from spacetable where substr(Date,0,5)='2015' and landing_outcome ='Failure (drone ship)'`
- Query result:

month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- I used "SELECT landing_outcome" and "COUNT(*), "WHERE", "AND", "GROUP BY" and "ORDER BY DESC" statements to rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Query: `%sql select landing_outcome, count(*) as Count from spacetable where date between '2010-06-04' and '2017-03-20' group by landing_outcome order by Count desc`
- Query result:

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

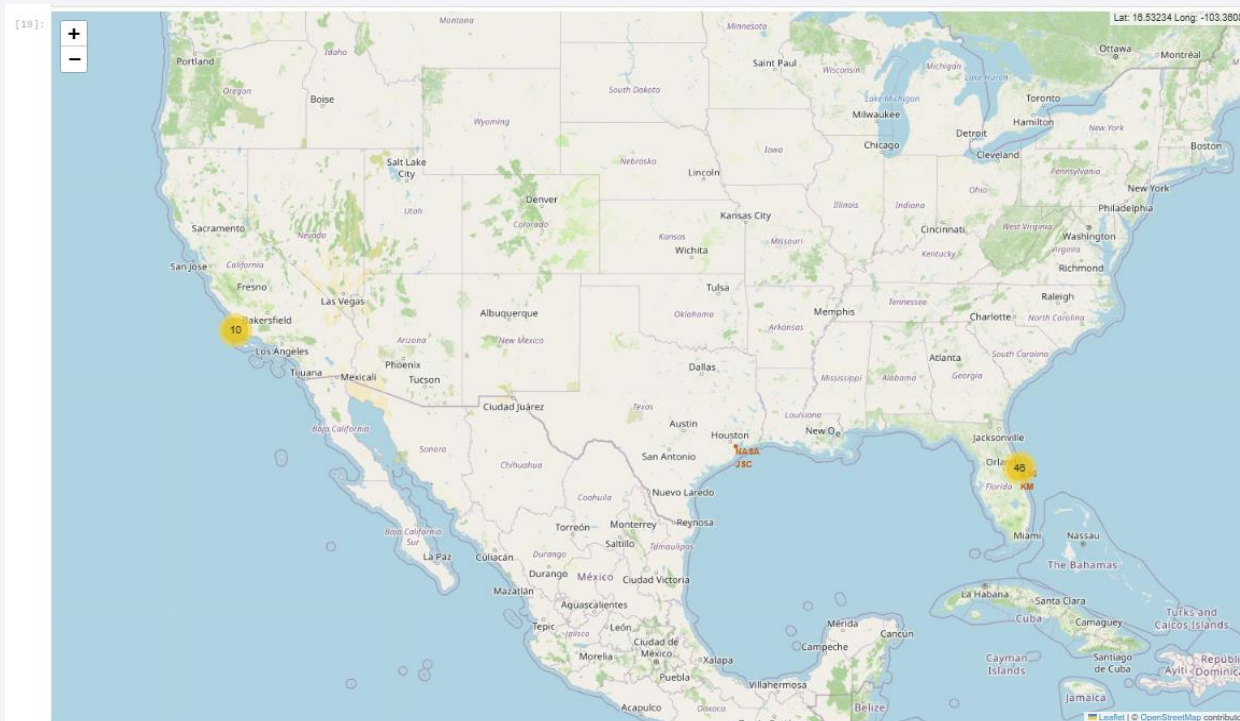
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

Section 3

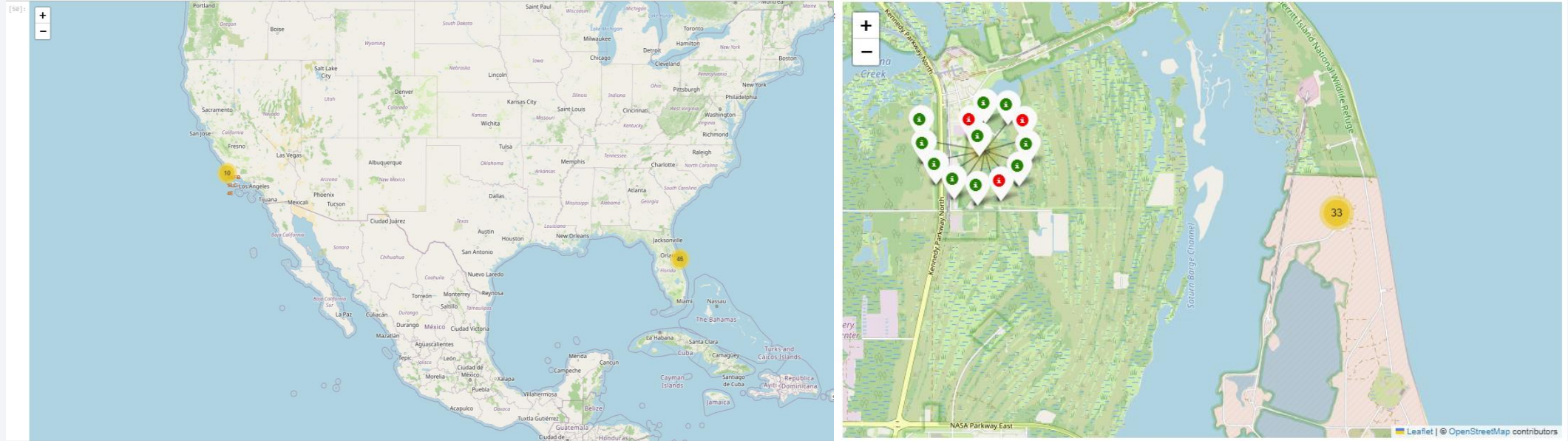
Launch Sites Proximities Analysis

All launch sites on a map

- The two launch site locations are VAFP SLC – 4E on the west coast of California The remaining 3 sites are on the eas coast of Florida.

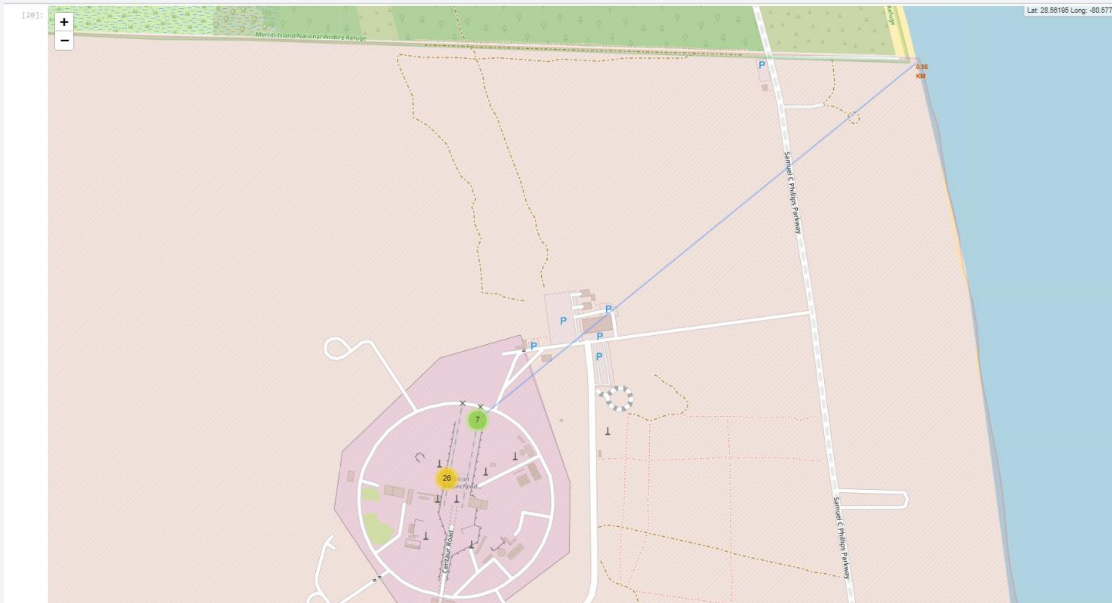


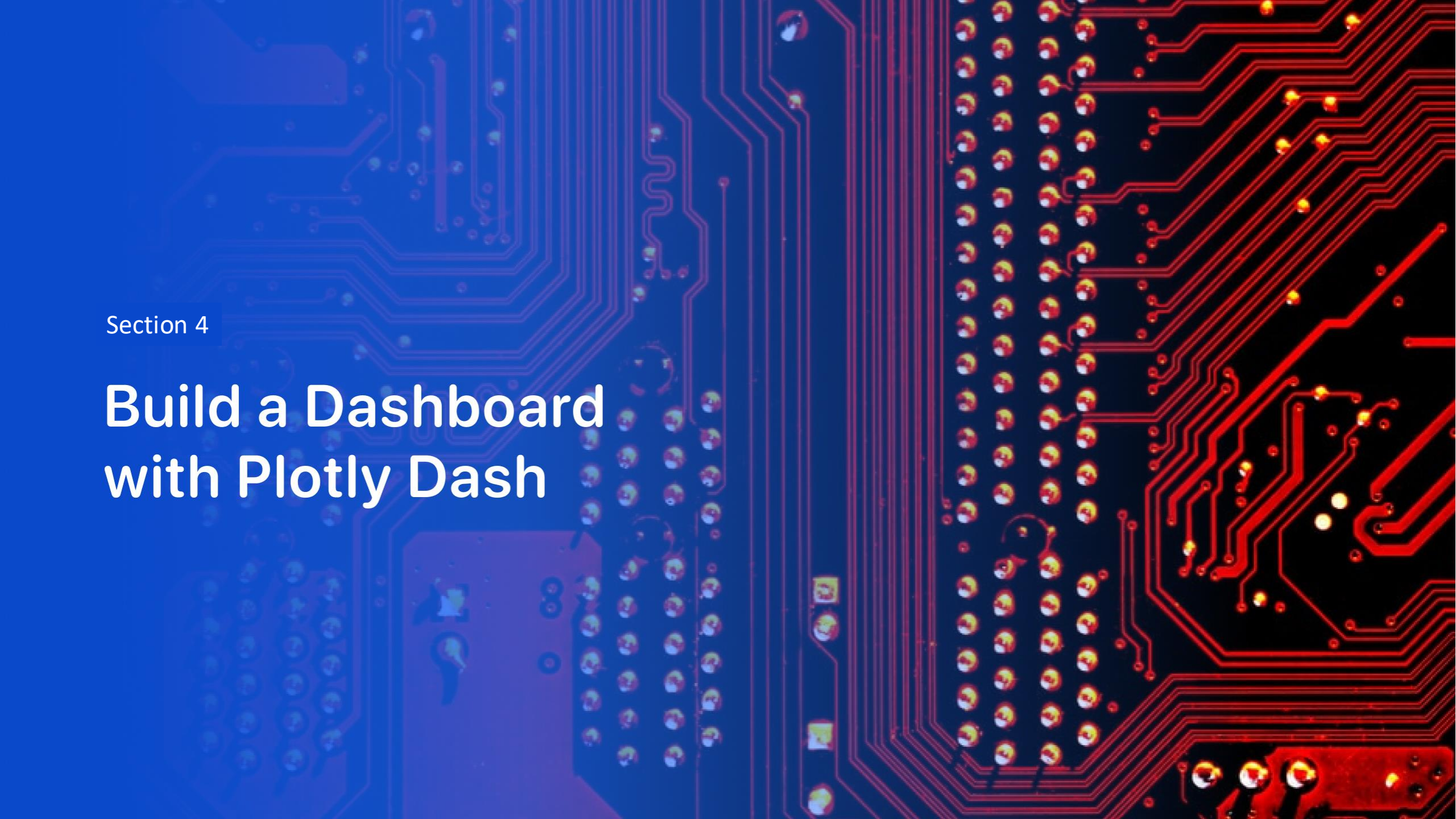
Success/failed launches for each site on the map



A launch site to its proximities

- CCAFS SLC-40 is 0.96 km where Merrit Island National Wildlife Refuge meets the coast (as shown in red, with the distance marker, on the map).



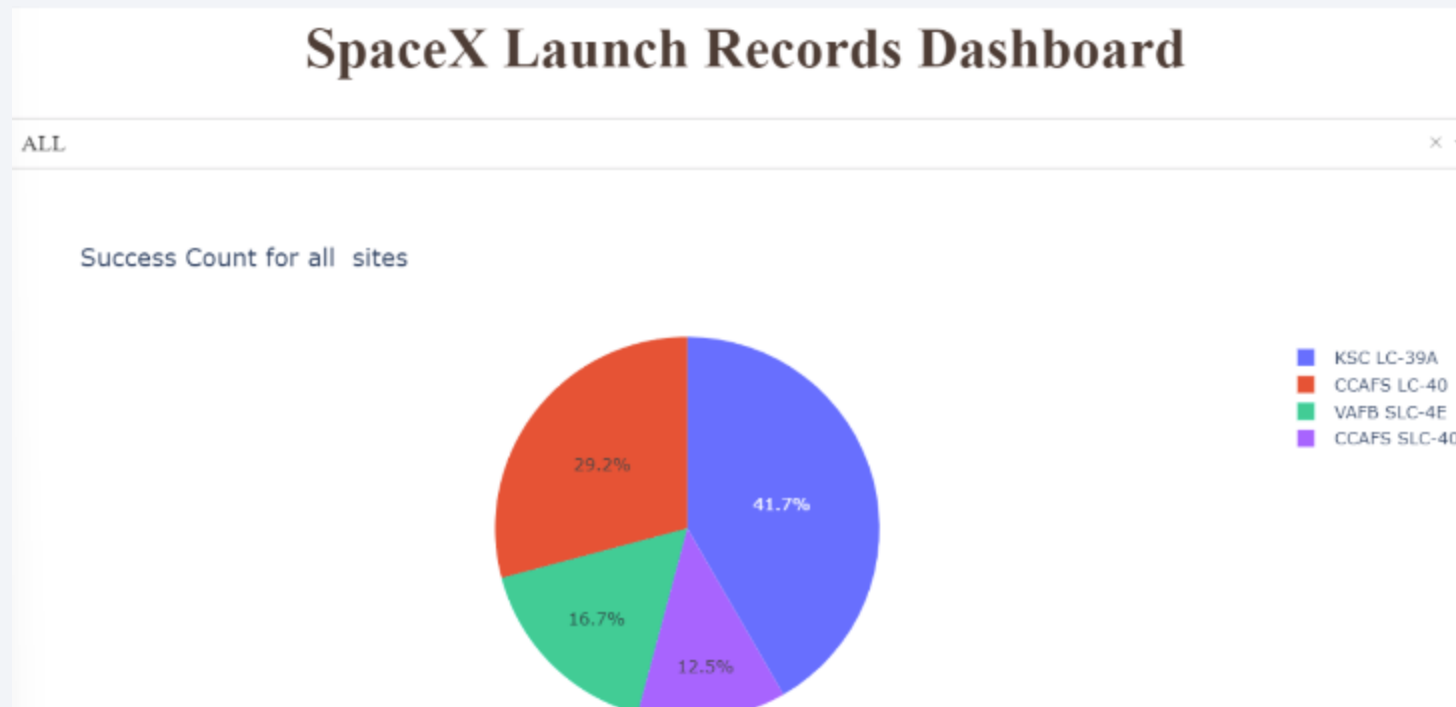


Section 4

Build a Dashboard with Plotly Dash

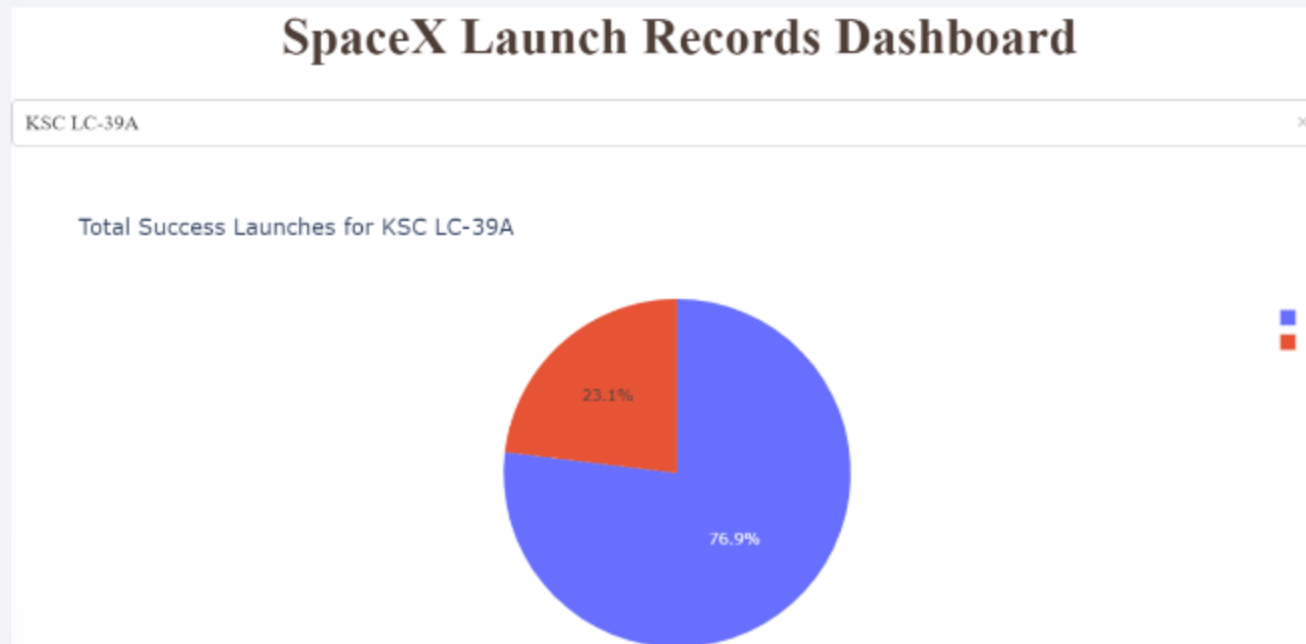
Launch success count for all sites

- Site KSC LC – 39A has the most successes



Launch site with highest launch success ratio

- The site had about 77% success rate and 23% failure rate



Payload vs. Launch Outcome Scatter Plots

- Left: < 5000 kg - Right; > 5000kg

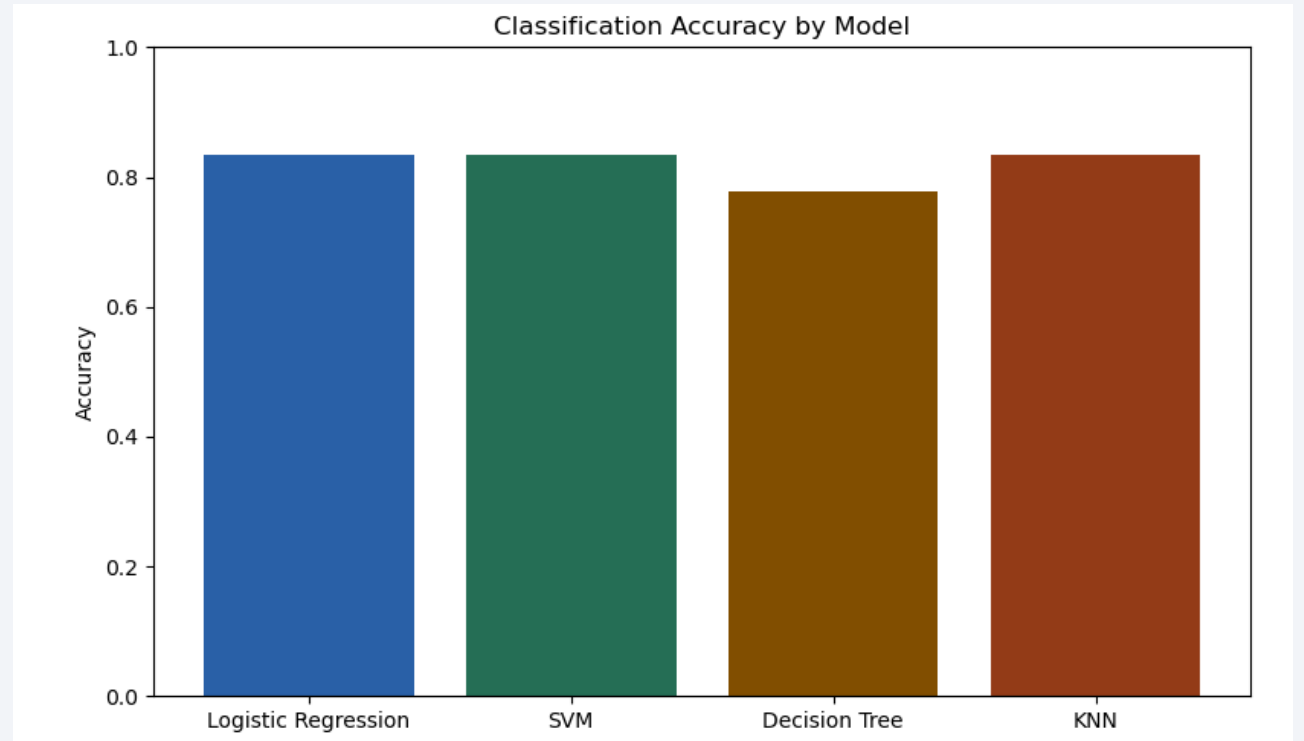


Section 5

Predictive Analysis (Classification)

Classification Accuracy

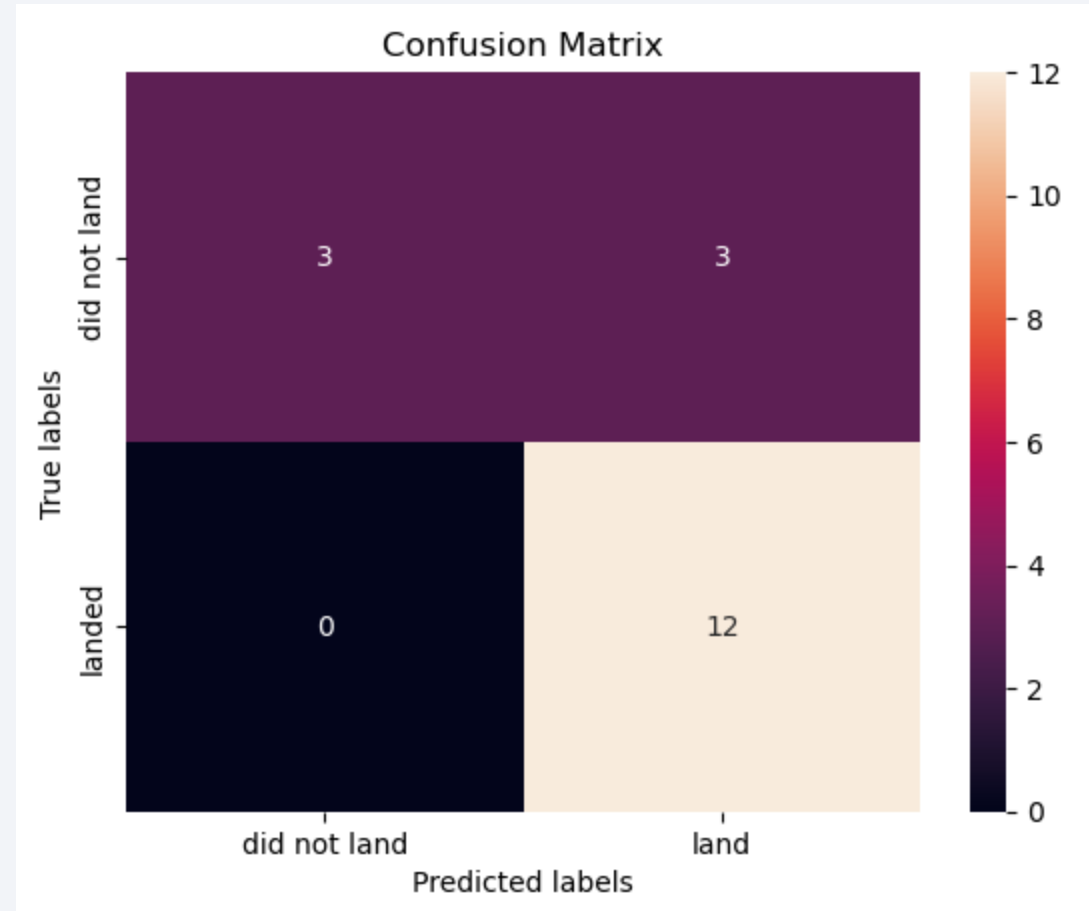
The best model is Logistic Regression because it is the simplest of the three.



Confusion Matrix

12 true positives, 3 false positives.

The problem with the model is that it sometimes predicts the rocket landed when it actually did not.



Conclusions

- The data were normalized using StandardScaler before training the models.
- The data were divided into 80% training and 20% testing (18 testing samples).
- Four classification algorithms were evaluated: Logistic Regression, SVM, Decision Tree, and KNN.
- The optimal hyperparameters were selected using GridSearchCV with 10-fold cross-validation.
- Logistic Regression, SVM, and KNN achieved the same accuracy in testing: 83.33%.
- Decision Tree was the worst-performing model: 77.78% accuracy.
- The main problem detected was false positives: the model predicted landings when none actually occurred.
- Logistic Regression is the recommended model because it is the simplest among those that achieved the best accuracy.

Appendix

- <https://github.com/mamemoverflow/testrepo>

Thank you!

